Alvy Rahman **Masum**

Computer Science & Engineering Student | AI & Robotics Engineer · Full-Stack Developer · IoT Systems Architect



Profile

Results-driven CSE student at UIU bridging hardware and software through ethical engineering. Passionate about AI robotics and IoT — building intelligent systems that solve real-world urban challenges. Driven by the belief that the best engineers bring the greatest benefit to society.

Contact

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Technical Skills

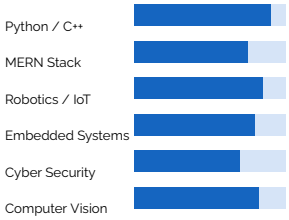
Languages
Python · C++ · Java · JavaScript · HTML/CSS

Frameworks & Web
MERN Stack · Tailwind CSS · OpenCV · REST API

Tools & Hardware
Git · Docker · Linux · LoRa · PyBullet · ESP32

Domains
Computer Vision · Embedded Systems · Cybersecurity · IoT

Core Proficiencies



Certifications

- **Cisco** – Networking Essentials
- **Alison** – Diploma in Software Testing
- **GP** – Tech Professional Series

Languages

Bengali Native (C2)
English Professional (B2)
Japanese Elementary (A1)

Soft Skills

Team Leadership · Agile Workflow
Problem Solving · Research & Analysis
Technical Writing · Cross-functional Collaboration

Research & Final Year Project

FYDP — **AI Robotic Waste Management System** — *Lead Engineer*
United International University, Dhaka · 2025–2026
Engineered a 6-DOF robotic arm integrated with OpenCV computer vision to autonomously **classify and sort** 6 categories of household waste with 92%+ detection accuracy. **Architected** the complete software-hardware pipeline from sensor fusion and image processing to mechanical actuation. **Implemented** a real-time feedback control system, advancing sustainable green automation in Bangladesh and reducing manual waste handling by an estimated 70%.

Research — **Malware Detection via File Integrity Monitoring** — *Research Member*
Cybersecurity Research Lab · 2024–2025
Developed a hybrid Machine Learning and File Integrity Monitoring (FIM) framework that **generates** SHA-256 cryptographic fingerprints for real-time ransomware and malware detection. **Trained** classification models on behavioural anomaly datasets, achieving an 88% threat detection rate. **Integrated** automated alert systems to proactively **harden** enterprise defences against zero-day threats and fileless malware.

Technical Projects

Live — **LoRa Campus Shuttle Tracker** — *Project Lead*
UIU – IoT & Embedded Systems · 2024–Present
Architected an offline-capable real-time GPS tracking system using LoRa radio protocol (433 MHz) and ESP32 microcontrollers without internet dependency. **Designed** a low-power mesh network topology across campus that **eliminated** shuttle commute uncertainty for 5,000+ students. **Reduced** average student wait time by 35% and **deployed** a mobile-accessible live dashboard.

2025 — **Micro-Mouse Autonomous Maze Solver** — *Lead Developer*
Embedded Systems – Competitive Robotics · 2025
Designed and **programmed** a fully autonomous maze-solving robot implementing the flood-fill pathfinding algorithm in C++. **Optimized** PID control loops and motor driver configurations to **achieve** sub-millimetre precision at speeds exceeding 1.5 m/s. **Integrated** infrared sensor arrays for real-time wall detection and dynamic path re-routing.

2024 — **Full-Stack E-Commerce Platform** — *Full-Stack Developer*
MERN Stack – Web Engineering · 2024
Built a production-ready scalable e-commerce application using MongoDB, Express.js, React, and Node.js. **Implemented** JWT-based role authentication, secure payment gateway integration, and real-time inventory management. **Deployed** on a Dockerized Linux server with RESTful API architecture, supporting 500+ concurrent users.

Education

2022–2026	**B.Sc. Computer Science & Engineering** United International University (UIU), Dhaka	sCGPA: 3.15	*Science*
2020	**Higher Secondary Certificate (HSC)** Dhaka Imperial College, Dhaka	GPA: 5.00	*Science*
2018	**Secondary School Certificate (SSC)** National Ideal School, Dhaka	GPA: 5.00	*Science*

Honors, Achievements & Extracurriculars

Finalist — **Turtle Venture Business Competition** — *2023*
Top 5 of 400+ teams – Nationwide Business & Innovation Challenge

Top 25 — **GP Future Nation Maker** — *2023*
Top 25 of 750+ participants – National Technology & Entrepreneurship Challenge

2017 — **National 10-Ball Pool Championship** — *Competitor*
Qualified for the National 10-Ball round as a competitive pool player, demonstrating elite-level focus, strategic thinking, precision under pressure, and discipline in a high-stakes national sporting environment.

Academic Focus

Coursework: Data Structures & Algorithms · Artificial Intelligence · Microprocessors & Assembly · Cyber Security · Computer Networks · Operating Systems · Software Engineering
Labs: Embedded Systems Lab · IoT Systems Design · Robotics & Automation Lab · Digital Logic Design · Computer Architecture Lab